

UCS658: Derivatives Pricing, Trading and Strategies

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Course Objective: Understanding methods of valuation and strategies of trading derivative instruments.

Syllabus

Law of One Price: Meaning, implication of the law of one price, no arbitrage model, usage in pricing of securities and derivative instruments

Pricing and Valuation: Basic principles, building blocks, assumptions, difference between price and value, pricing and valuation of basic derivative instruments

Basics of Option Pricing: Meaning of options, types of options, difference between options and basic derivative instruments

Laboratory Work: To use the proprietary software in live international derivatives

1. Introduction to derivatives strategies
2. Introduction to an additional international derivatives product
3. Learning and creating trading strategies
4. Practicing the strategies on the proprietary software
5. Understanding the role of derivatives in risk reduction

Recommended Prerequisites: Courses Finance, Accounting and Valuation + Financial Markets and Portfolio Theory.

Course Learning Objectives (CLO)

After the completion of the course, the student will be able to:

1. Explain and apply the law of one price in pricing models
2. Explicate and apply formulae to calculate price and value of a generic forward contract
3. Explain and apply formulae to calculate price and value of basic derivative instruments
4. Explicate the meaning and types of options
5. Apply the basic methods to calculate option prices

Reference Books:

1. Fundamentals of Futures and Options Markets - John C. Hull
2. Ross, Westerfield, Jaffe & Jordan. Corporate Finance: Core Principles and Applications.